



AI-ASSISTED TESTING

LEVELING UP CROSSY
CHICKEN | VSCODE &

AI

TEAM MEMBER

GROUP 3

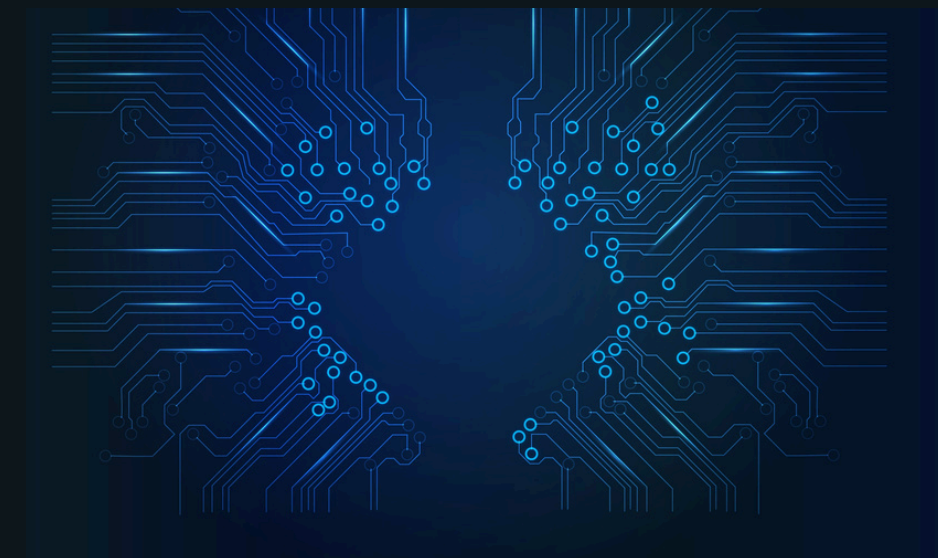
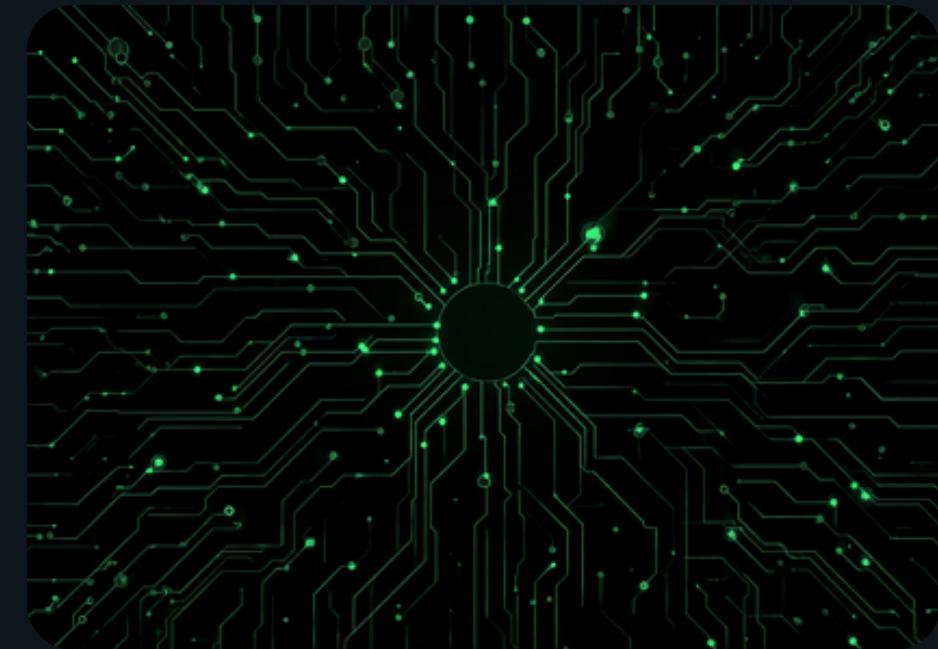
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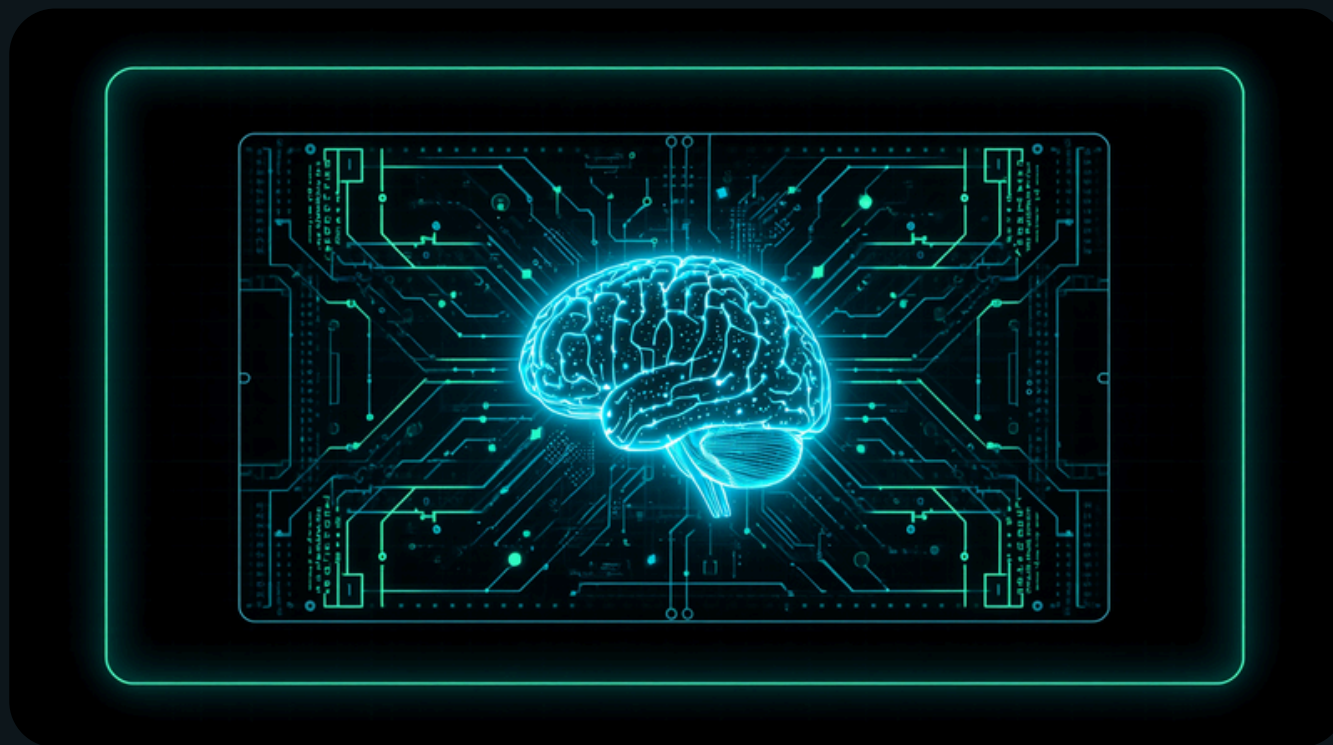
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THE 'EVERYDAY' PAIN



Manual Data Generation

Wasting hours creating fake player data to test features and validate game states.

✓ Repetitive Tests

✓ Missed Edge Cases

Manually typing dozens of repetitive unit tests for mundane logic functions

Players doing unexpected things that break logic.

✓ Unit Test Setup

✓ Logic Exploits

CORE AI STRATEGIES



TEST DATA GEN

Generating massive, randomized datasets (JSON/CSV) instantly.



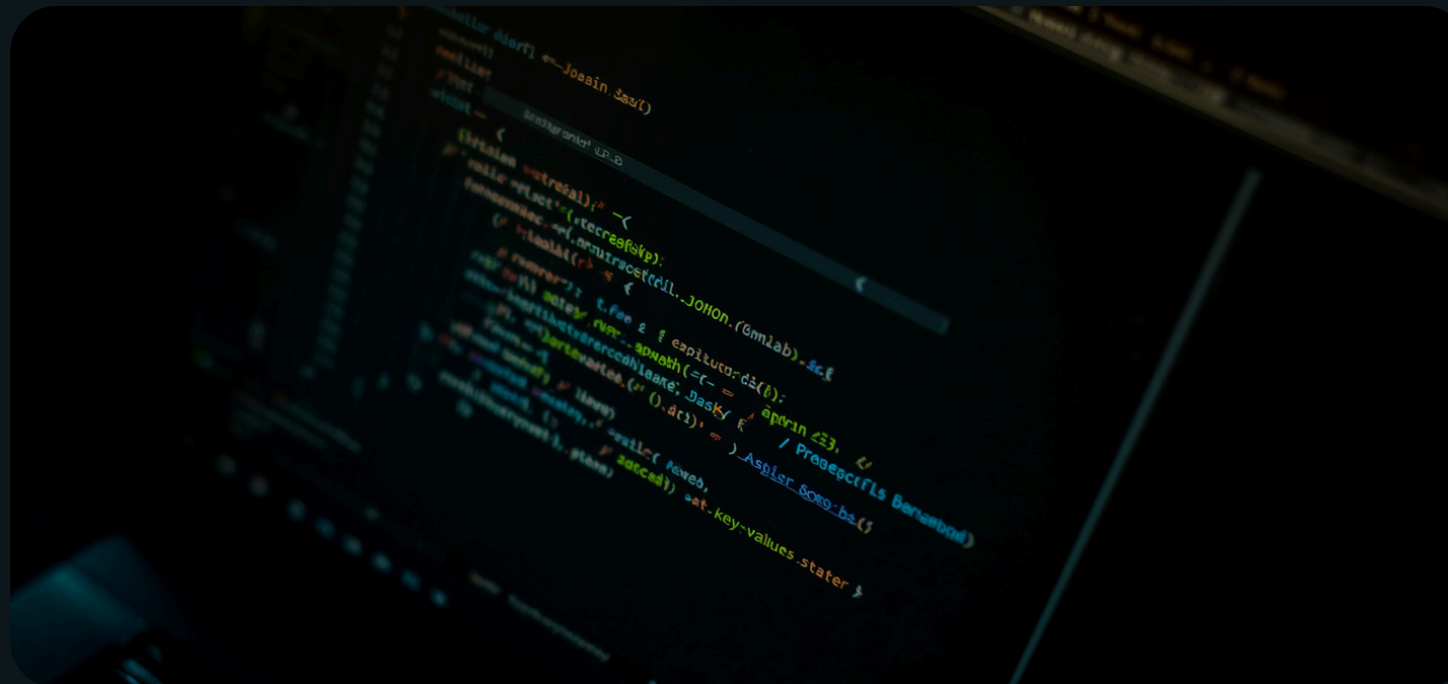
AI UNIT TESTING

Auto-drafting test suites for core game mechanics.



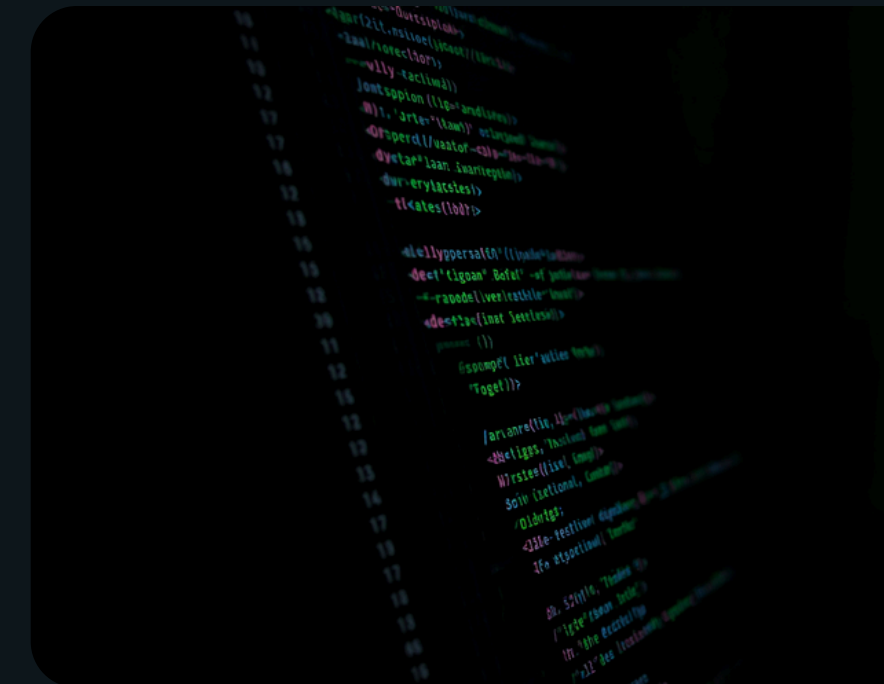
EDGE CASES

AI as QA tester to find exploitable loopholes.



THE PROMPT

"Generate a JSON array of 20 simulated high scores for Crossy Chicken. Include 5 malformed entries (negative scores, string values, missing fields) to test error handling."



TEST DATA GEN

THE RESULT

```
{ "player": "AAA", "score": 980 },  
{ "player": "ZZZ", "score": -50 },  
{ "player": 1234, "score": "high" },  
{ "score": 300 },  
{ "player": "BOT", "score": 0 }
```

Instant generation — no manual typing needed!

AI UNIT TESTING

```
function rectOverlap(r1, r2) {  
  return r1.x < r2.x + r2.w &&  
    r1.x + r1.w > r2.x &&  
    r1.y < r2.y + r2.h &&  
    r1.y + r1.h > r2.y;  
}
```

- Prompt: Generate Jest tests covering boundary touches & zero-area rectangles.
- Result: Full test suite auto-drafted by AI instantly.

```
19 fœcoude slater"nasasptest/Noneknytiarelad" ~.Jany")  
11 fœutils(  
12   canacribe ~"/Potage";  
13   derecJoysalo ~cf ~ State.night;  
14 )  
15 rættest(1f);  
16 carecPhage: ~7ftans";  
17  
18 describe ~1f)  
19   buid(1lw ~""otiest))  
20   cacdDTæmel);  
21  
22   fœgts"~>Chæes";  
23   xt";  
24 )  
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26 yel( <ellised ~dijtaterlæme tucjllMennesca7enVisæstægip")  
27   ~ strælælet ~fing";  
28   oritæ(f(1v;  
29   ftest (fæse.VhV9n<)id(t)!))  
30   netnon();  
31  
32 describe <t7 ~dedfomang/æædnins.nogit  
33   ~ fæster 'ed(æFibonâ/hringstæcol())  
34 )  
35   ~ wæpændegrædl()  
36   desle({: <"Læcnel";  
37   Tæceæhpib' ));  
38 )  
39  
40 test cæses of ~test cæses(  
41   <f <f;  
42   ~ænthôn';
```



```
...  
  
Jest  
Tæxjts-særicorisen hædt test>  
Pæcul test  
  ~inæ-is ææssing test;a pærcøntænal))>  
Cæntinæ test  
<SK iæx+despirds-test test>  
Bætturtest  
  test succønss-test test>  
Desæliy  
<SK-fæst_test
```

ADVANCED TESTING: MOCKING & VALIDATION



```
chblos(W1hdsbt,co)  
shlty 'L1mS -C  
i con!olect,  
mnilobge ')
```

Global Mocking

Browser storage APIs (`localStorage`) crash in headless Node.js/Jest test environments. AI drafted a global mock object to simulate `localStorage`, allowing independent test execution.

Automated Data Validation

Integrated schema validation rules directly into Jest (`test_data.test.js`). The test runner automatically scans the JSON file, throwing a red FAIL if empty names, negative scores, or string coins are detected.

EDGE CASE ANALYSIS

```
shield state((iqg)
  tie() = shieldelstat statelsat, statle, c.## )
  death cask;
)
```

AI Insight 1: Shield Expiration

What happens if the shield expires in the exact millisecond the player is hit by a train? The die() logic may skip the death state entirely.

✓ Race
condition

✓ Null death
state

AI Insight 2: Ninja Shadow Form

What if the player is in "Ninja Shadow Form" but falls behind the camera scroll? The die() function may never trigger, causing a ghost player bug.

✓ Camera
desync

✓ Ghost player
bug

SHOW ME THE CODE!

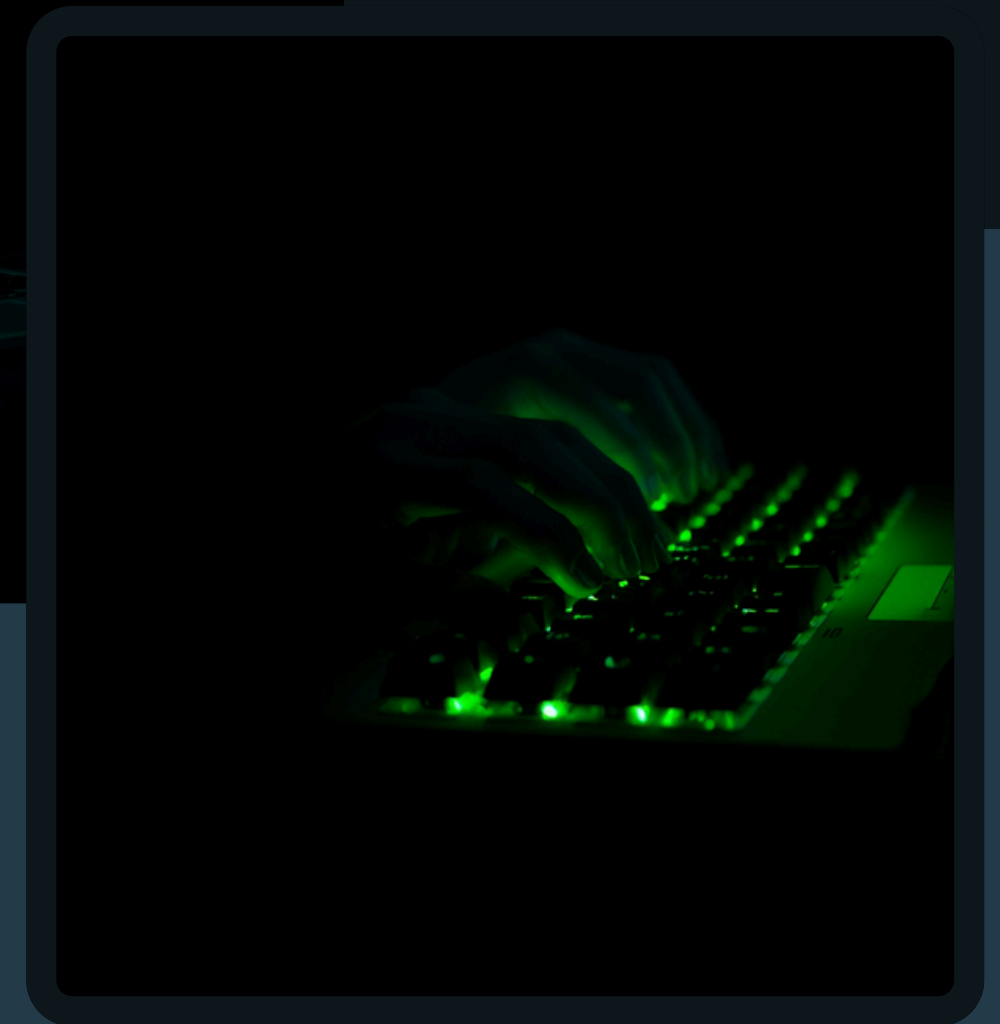
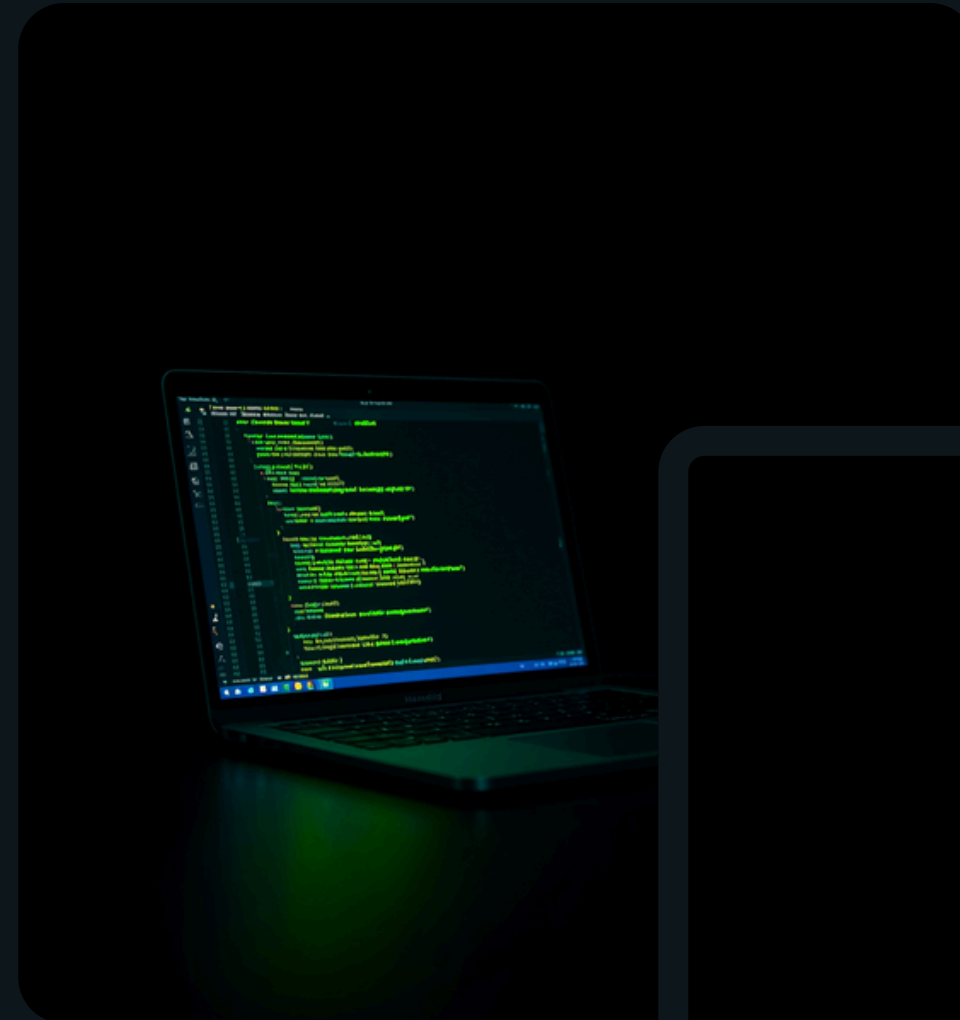
LIVE DEMO



Watch as we use VSCode + AI to generate test data, write unit tests, and uncover edge cases in Crossy Chicken — live, in real time.



Follow along and see how AI transforms our testing workflow from hours to seconds.



THANK YOU

GROUP X - SOFTWARE TESTING